Semantic-Driven Multi-UAV Cooperative Communication With Joint Positioning and Resource Allocation

Wanli Wen, Weiran Guo, Liang Liang, and Yunjian Jia, *Member, IEEE*

***Abstract*—Semantic communication represents a new paradigm for sixth-generation wireless networks that prioritizes meaning-centric transmission. Uncrewed aerial vehicles (UAVs) provide deployment flexibility and line-of-sight connectivity for coverage. The integration of these technologies presents challenges in the optimization of UAV deployment, radio resources, and semantic content fidelity. This letter proposes a framework for multi-UAV-enabled semantic communication that uses coordinated multipoint transmission. We formulate a problem to maximize the sum Quality of Experience (QoE) of all users by optimizing UAV positioning, cooperative beamforming, and semantic compression ratios, which results in a mixed-integer nonlinear programming problem. To solve this problem, we develop a hybrid alternating learning and optimization (HALO) algorithm. HALO decomposes the problem, using proximal policy optimization for discrete semantic compression decisions and an alternating optimization method for the continuous UAV positioning and beamforming variables. Simulation results show that the proposed framework achieves QoE improvements over benchmark methods, which validates the benefits of joint optimization in semantic-aware UAV networks.**

***Index Terms*—Semantic communication, UAVs, cooperative communication, resource allocation, proximal policy optimization.**

I. Introduction

SEMANTIC communication, a paradigm for sixth-generation networks, transmits information based on meaning rather than bit-rates [1]. This paradigm shift is essential for supporting delay-sensitive and data-intensive 6G applications such as immersive communication and edge AI collaboration. These demanding scenarios require high semantic fidelity with minimal transmission overhead. The performance of such systems depends on channel quality. Uncrewed aerial vehicles (UAVs) provide flexible, on-demand coverage and line-of-sight (LoS) links, making them ideal for dynamic semantic services where meaning-centric delivery maximizes data utility over constrained wireless channels [2], [3]. The integration of these technologies requires a method for managing UAV mobility and coordinating radio resources in consideration of semantic information, which forms a resource allocation problem.

Received 1 December 2025; accepted 1 January 2026. Date of current version 13 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62571071 and Grant 62571072; in part by the Natural Science Foundation of Chongqing under Grant CSTB2023NSCQ-LZX0079; and in part by the Fundamental Research Funds for the Central Universities under Grant 2023CDJKYJH014. The associate editor coordinating the review of this article and approving it for publication was C. Y. Leow. *(Corresponding author: Wanli Wen.)*

The authors are with the School of Microelectronics and Communication Engineering, Chongqing University, Chongqing 400044, China (e-mail: wanli_wen@cqu.edu.cn; weiran_guo@stu.cqu.edu.cn; liangliang@cqu.edu.cn; yunjian@cqu.edu.cn).

Digital Object Identifier 10.1109/LWC.2026.3651641

Existing studies related to this letter can be broadly grouped into three categories. The first category focuses on semantic communication design under largely static topologies, without exploiting the additional degrees of freedom offered by UAV mobility [1], [4], [5], [6]. The second category considers UAV trajectory design, beamforming, or power allocation using optimization or DRL techniques, but remains semantic-agnostic and optimizes traditional metrics such as throughput [7], [8], energy efficiency [9], [10], or classical QoE [2], [11]. The third category begins to combine semantic communication with mobile platforms but typically optimizes either the semantic policy or the UAV control in isolation [12]. A framework is therefore needed to connect the semantic layer with a physical layer that manages UAV positioning and cooperative transmission. For a multi-UAV system, such cooperative transmission is formally realized through coordinated multipoint (CoMP) techniques, which necessitates the joint design of beamforming across the fleet.

This letter presents a framework for multi-UAV cooperative semantic communication. Our contributions are as follows:

1) We model a system that integrates a multi-UAV CoMP physical layer with a task-oriented semantic communication layer. The model clarifies the relationship between UAV mobility, radio resources, and semantic fidelity.
2) We formulate a problem to maximize the sum QoE of all users by jointly optimizing UAV positioning, cooperative beamforming, and semantic compression ratios, which is a mixed-integer nonlinear programming (MINLP) problem.
3) We develop a hybrid alternating learning and optimization (HALO) algorithm to solve the formulated MINLP. Unlike conventional single-layer or single-tool approaches, the proposed HALO framework forms a tightly coupled learning optimization loop, where a PPO-based semantic policy and an alternating optimization (AO) method iteratively refine each other to handle the mixed discrete-continuous structure of semantic compression, UAV positioning, and cooperative beamforming.

II. System Model

We consider a downlink multi-UAV cooperative semantic communication system, depicted in Fig. 1. The network architecture is engineered to provide on-demand, high-quality multimedia services to a set of ground users by leveraging the principles of semantic communication. The system operates over a sequence of discrete time slots, indexed by $t \in \{1, 2, \dots\}$, allowing for dynamic adaptation of UAV positions and resource allocation strategies. The network consists of two principal classes of agents: a cooperative fleet of N rotary-wing UAVs, indexed by $\mathcal{N} \triangleq \{1, 2, \dots, N\}$, and K

2162-2345 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Chung-ang Univ. Downloaded on October 27,2026 at 10:36:29 UTC from IEEE Xplore. Restrictions apply.

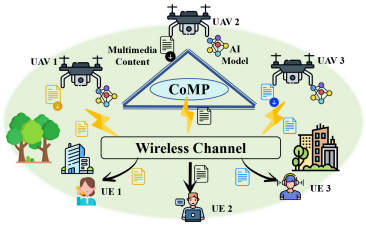


Fig. 1. System model.

ground users, indexed by $\mathcal{K} \triangleq \{1, 2, \ldots, K\}$. Each UAV is provisioned with the designated multimedia content and a powerful pre-trained artificial intelligence (AI) model engineered for end-to-end semantic transmission. This empowers the UAVs, acting as intelligent aerial base stations, to transcend mere data relaying and instead transmit only the semantically pertinent information to their respective users. The operational environment is modeled within a three-dimensional Cartesian coordinate system, where ground users $k \in \mathcal{K}$ possess fixed positions, denoted by $\mathbf{u}_k \in \mathbb{R}^2$. In each time slot t, UAV $n \in \mathcal{N}$ dynamically adjusts its position, while maintaining a constant altitude H. A cornerstone of our physical layer design is CoMP transmission, under which the entire UAV fleet functions as a distributed multi-antenna transmitter. This cooperative strategy is essential to generate the high-quality channels required to support reliable semantic information delivery, facilitating coherent signal combination and sophisticated interference management across the network.

A. Semantic Communication Model

The core of our semantic framework is a deep neural network trained end-to-end on a large-scale, shared knowledge base [1]. This model learns a direct mapping from the source data to channel-resilient symbols, effectively unifying the tasks of semantic feature extraction and robust channel encoding within a single architecture. The original data intended for user k, denoted by $\mathbf{s}_k$, is thus processed by the learned semantic encoder onboard the UAVs to extract its essential meaning.

The level of information abstraction is governed by a key parameter, the semantic compression ratio (SCR), denoted by τ_k^t. We model the SCR as a discrete variable for practical implementation. A lower value of τ_k^t corresponds to a more detailed representation (less compression) and thus higher potential semantic fidelity, albeit at the cost of increased channel resource consumption. We constrain the SCR to a set of Γ predefined levels

$$\tau_k^t \in \{1, 2, \ldots, \Gamma\}, \quad \forall k \in \mathcal{K}. \tag{1}$$

The vector of SCR decisions for all users at time slot t is denoted by $\boldsymbol{\tau}^t \triangleq (\tau_k^t)_{k\in\mathcal{K}}$.

B. UAV Deployment and Mobility Model

The efficacy of the UAV-aided network is fundamentally determined by the strategic deployment of its aerial platforms. The mobility of the N UAVs is governed by a set of operational constraints that ensure mission safety and physical feasibility. We use $\mathbf{q}_n^t \in \mathbb{R}^2$ to denote the horizontal position coordinates of UAV n at time t.

First, the UAVs are confined to operate within a designated horizontal service area, $\mathcal{A} \subset \mathbb{R}^2$, ensuring they remain over the intended coverage region:

$$\mathbf{q}_n^t \in \mathcal{A}, \quad \forall n \in \mathcal{N}. \tag{2}$$

Second, to guarantee safe operation and prevent mid-air collisions, a minimum separation distance, $d_{\min}$, must be maintained between the horizontal positions of any two UAVs. Thus, we have

$$\|\mathbf{q}_n^t - \mathbf{q}_m^t\|_2 \geq d_{\min}, \quad \forall n, m \in \mathcal{N}, n \neq m. \tag{3}$$

Finally, the kinematic limitations of the UAVs are modeled by bounding their maximum speed, $v_{\max}$. The displacement of any UAV during a time slot of duration Δt is thereby constrained by

$$\|\mathbf{q}_n^t - \mathbf{q}_n^{t-1}\|_2 \leq v_{\max}\Delta t, \quad \forall n \in \mathcal{N}. \tag{4}$$

Collectively, the above constraints define the feasible positioning space for the UAV fleet. The vector of all UAV positions is $\mathbf{q}^t \triangleq (\mathbf{q}_n^t)_{n\in\mathcal{N}}$.

C. Cooperative Transmission Model

Under the CoMP framework, all N UAVs cooperatively transmit signals to the K users. Each UAV is equipped with a uniform linear array of M antennas, while each ground user possesses a single antenna. In time slot t, the beamforming vector transmitted from UAV n to user k is denoted by $\mathbf{w}_{n,k}^t \in \mathbb{C}^{M\times 1}$. The transmit power of each UAV n is limited by its maximum capability, P_n, such that

$$\sum_{k\in\mathcal{K}} \|\mathbf{w}_{n,k}^t\|_2^2 \leq P_n, \quad \forall n \in \mathcal{N}. \tag{5}$$

The air-to-ground (A2G) channel from UAV n to user k, $\mathbf{h}_{n,k}^t \in \mathbb{C}^{M\times 1}$, is characterized by both large-scale path loss and small-scale fading. We model this channel as $\mathbf{h}_{n,k}^t = \sqrt{\beta_{n,k}^t}\mathbf{g}_{n,k}^t$, where $\mathbf{g}_{n,k}^t \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_M)$ represents the small-scale Rayleigh fading component. The term $\beta_{n,k}^t$ is the large-scale path loss, which is fundamentally dependent on the distance between the UAV and the user. A widely adopted model for path loss is given by $\beta_{n,k}^t = \beta_0(d_{n,k}^t)^{-\alpha}$, where β_0 is the channel power gain at a reference distance of one meter, and $\alpha \geq 2$ is the path loss exponent. The 3D Euclidean distance between UAV n and user k at time t, $d_{n,k}^t$, is calculated as $d_{n,k}^t = \sqrt{\|\mathbf{q}_n^t - \mathbf{u}_k\|_2^2 + H^2}$.

The received signal-to-interference-plus-noise ratio (SINR) at user k is a direct consequence of the cooperative transmission and is given by $\gamma_k = \frac{|\sum_{n\in\mathcal{N}}(\mathbf{h}_{n,k}^t)^{\mathrm{H}}\mathbf{w}_{n,k}^t|^2}{\sum_{i\in\mathcal{K}\setminus\{k\}}|\sum_{n\in\mathcal{N}}(\mathbf{h}_{n,k}^t)^{\mathrm{H}}\mathbf{w}_{n,i}^t|^2+\sigma^2}$, where $(\cdot)^{\mathrm{H}}$ denotes the conjugate transpose, and σ^2 is the additive white Gaussian noise power. To ensure a baseline of reliable decoding, the SINR for each user must satisfy a minimum threshold, γ_{th}. So, we have the following constraint:

$$\gamma_k \geq \gamma_{\mathrm{th}}, \quad \forall k \in \mathcal{K}. \tag{6}$$

D. QoE Model

In semantic communication, the ultimate performance metric is not the raw data rate but the fidelity of the conveyed meaning. We quantify this fidelity using semantic similarity, $\varepsilon_k^t \in [0, 1]$, which measures the cosine similarity between the high-dimensional feature embeddings of the original source data, $\mathbf{s}_k$, and the reconstructed data, $\hat{\mathbf{s}}_k^t$. For a concrete

Authorized licensed use limited to: Chung-ang Univ. Downloaded on October 27,2026 at 10:36:29 UTC from IEEE Xplore. Restrictions apply.

implementation, we consider the AI model for semantic processing to be a Transformer-based architecture [5], where the embedding function $\mathrm{B}(\cdot)$ maps text or image data to a semantic vector space. The similarity is then calculated as $\varepsilon_k^t = \frac{\mathrm{B}(\mathbf{s}_k)\mathrm{B}(\hat{\mathbf{s}}_k^t)^{\mathrm{T}}}{\|\mathrm{B}(\mathbf{s}_k)\|_2\|\mathrm{B}(\hat{\mathbf{s}}_k^t)\|_2}$. To guarantee a minimum level of meaningful communication, this semantic similarity must be maintained above a predefined threshold, $\varepsilon_{\mathrm{th}}$, i.e.,

$$\varepsilon_k^t \geq \varepsilon_{\mathrm{th}}, \quad \forall k \in \mathcal{K}. \tag{7}$$

We adopt QoE as our primary optimization objective, as it directly reflects user satisfaction by capturing both the quantity and quality of the transmitted semantic information. Drawing upon established principles in multimedia streaming that model perceived quality logarithmically [11], we formulate the QoE for user k in slot t as $\mathrm{QoE}_k^t = \lambda_1 \log(1+\lambda_2 \frac{I_k \varepsilon_k^t}{L_k \tau_k^t})$. Here, the term $\frac{I_k \varepsilon_k^t}{L_k \tau_k^t}$ can be interpreted as the effective semantic rate, a measure of the useful information successfully delivered, where I_k is the amount of semantic information per sentence and L_k is the number of words per sentence. The parameters λ_1 and λ_2 are scaling constants that map this effective rate to a numerical satisfaction score [11]. This logarithmic formulation models the diminishing returns of user satisfaction with respect to increases in the semantic rate, thus providing a robust and meaningful objective for our problem.

In contrast to traditional objectives that vary monotonically with SINR or throughput, our QoE formulation is jointly influenced by the semantic similarity ε_k^t and the SCR τ_k^t. The similarity ε_k^t, generated by the pre-trained DeepSC-VQA model [4], is not a linear or monotonic function of γ_k, while τ_k^t simultaneously shapes the logistic mapping between γ_k and ε_k^t and determines the effective semantic rate. As a result, increasing SINR alone does not necessarily yield higher QoE, and the optimization must balance semantic fidelity and resource expenditure through an appropriate SCR policy.

III. Problem Formulation and Transformation

Building upon the user-centric and semantic-driven QoE model derived in Section II, our objective is to maximize the sum-QoE by jointly optimizing the UAV positions and associated resources. Here, resources include both radio-layer assets (cooperative beamformers) and semantic-layer assets (SCRs). For each time slot t, this translates to optimizing the UAV positions $\mathbf{q}^t$, the beamforming vectors $\mathbf{w}^t$, and the SCR assignments $\boldsymbol{\tau}^t$. The problem is formally stated as

Problem 1 (Sum-QoE Maximization Problem):

$$\begin{aligned} \max_{\mathbf{q}^t, \mathbf{w}^t, \boldsymbol{\tau}^t} \quad & \sum_{k \in \mathcal{K}} \mathrm{QoE}_k^t \\ \text{s.t.} \quad & (1),\ (2),\ (3),\ (4),\ (5),\ (6),\ (7). \end{aligned}$$

Problem 1 is a MINLP problem. The key challenge in solving this problem arises from the semantic similarity function ε_k^t. This function lacks a closed-form analytical expression with respect to the optimization variables $(\mathbf{q}^t, \mathbf{w}^t, \boldsymbol{\tau}^t)$, rendering direct optimization impossible. The value of ε_k^t is the output of the complex, non-linear AI model, which acts as a *closed box* within the optimization framework.

To render Problem 1 tractable, we adopt a standard methodology from the literature [6] and approximate the performance of the semantic communication system. We model the relationship between the semantic similarity ε_k^t and its key input drivers, i.e., the SINR γ_k and the SCR τ_k^t, using a logistic function. This function effectively captures the characteristic S-shaped performance curve of communication systems, where accuracy saturates at high SINR. To be specific, the approximated similarity, $\tilde{\varepsilon}_k^t$, can be expressed as $\varepsilon_k^t \approx \tilde{\varepsilon}_k^t \triangleq \frac{c_{\tau_k^t}}{1+e^{-a_{\tau_k^t}(\gamma_k - b_{\tau_k^t})}}$, where $a_{\tau_k^t}$, $b_{\tau_k^t}$, and $c_{\tau_k^t}$ are fitting parameters that characterize the performance of the given AI model at each SCR level τ_k^t. These parameters are determined via offline empirical measurements. By substituting the tractable approximation into Problem 1, we arrive at an approximated, yet still challenging, optimization problem, i.e., Problem 2.

Problem 2 (Approximation of Problem 1):

$$\begin{aligned} \max_{\mathbf{q}^t, \mathbf{w}^t, \boldsymbol{\tau}^t} \quad & \sum_{k \in \mathcal{K}} \lambda_1 \log\left(1 + \lambda_2 \frac{I_k \tilde{\varepsilon}_k^t}{L_k \tau_k^t}\right) \\ \text{s.t.} \quad & \tilde{\varepsilon}_k^t \geq \varepsilon_{\mathrm{th}}, \quad \forall k \in \mathcal{K}, \\ & (1),\ (2),\ (3),\ (4),\ (5),\ (6). \end{aligned} \tag{8}$$

Despite this reformulation, Problem 2 remains a MINLP due to three fundamental issues: (i) the strong coupling between the continuous variables $\mathbf{q}^t$ and $\mathbf{w}^t$ within the non-convex SINR and QoE expressions; (ii) the combinatorial nature of the discrete SCR variable $\boldsymbol{\tau}^t$; and (iii) the non-convexity of the collision avoidance constraint in (3). An exhaustive search over the discrete variables would lead to exponential complexity, rendering it impractical for real-time applications. This complex structure necessitates the development of a sophisticated and efficient optimization algorithm.

IV. The Proposed HALO Algorithm

To solve Problem 2, we propose the HALO algorithm, which decomposes the problem into its discrete and continuous domains. Specifically, we employ a DRL agent to manage the combinatorial selection of discrete SCRs, while a classical optimization method is used to handle the continuous variables of UAV positioning and beamforming.

A. Discrete SCR Selection via PPO

We model the SCR vector $\boldsymbol{\tau}^t$ selection as a sequential decision process and solve it using a centralized DRL agent. This centralized control is essential for realizing CoMP joint beamforming and ensuring the global coordination of discrete SCR selection to maximize the sum QoE. The Markov Decision Process (MDP) is defined as follows:

State(s_k^t)**:** The state for making a decision for user k includes the full channel state information (CSI) matrix, $\mathbf{H}^t = \{\mathbf{h}_{n,j}^t\}_{\forall n,j}$, and the SCRs assigned to the preceding $k-1$ users.

Action(a_k^t)**:** The action is the selection of an SCR level for user k, i.e., $a_k^t = \tau_k^t \in \{1, 2, \ldots, \Gamma\}$.

Reward(r_k^t)**:** The immediate reward is the QoE achieved for user k after the subsequent positioning and beamforming

Authorized licensed use limited to: Chung-ang Univ. Downloaded on October 27,2026 at 10:36:29 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 The Proposed HALO Algorithm

PPO Initialization: Initialize actor network π_θ and critic network V_ϕ.
for episode $= 1, 2, \ldots, E_{\max}$ **do**
  Observe initial state s_1.
  Initialize empty positioning buffer $\mathcal{D}$.
  for user $k = 1, \ldots, K$ **do**
    Agent selects SCR $\tau_k^t \sim \pi_\theta(\cdot|s_k)$.
    Obtain $\mathbf{q}^{t*}, \mathbf{w}^{t*}$ by solving the subproblem in Section IV-B given $\boldsymbol{\tau}^t$.
    Calculate reward $r_k^t = \mathrm{QoE}_k(\mathbf{q}^{t*}, \mathbf{w}^{t*}, \tau_k^t)$.
    Observe s_{k+1}; store $(s_k, \tau_k^t, r_k^t, s_{k+1})$ in $\mathcal{D}$.
  end for
  Update π_θ and V_ϕ using PPO updates with $\mathcal{D}$.
end for

subproblem is solved. This directly connects the agent's action to the overarching system objective.

For the DRL algorithm, we employ PPO, an actor-critic method renowned for its stability and data efficiency in Algorithm 1. The actor network, π_θ, maps states to a probability distribution over actions, while the critic network, V_ϕ, estimates the state-value function. PPO optimizes a clipped surrogate objective function to ensure stable policy updates.

B. Joint Positioning and Beamforming via AO

For a given SCR vector $\boldsymbol{\tau}^t$ determined by the PPO agent, Problem 2 simplifies into a continuous optimization subproblem over variables $(\mathbf{q}^t, \mathbf{w}^t)$. Due to the coupling between these variables, we propose an efficient alternating optimization approach in Algorithm 2, which iteratively optimizes one set of variables while keeping the other fixed until convergence.

1) Beamforming Optimization for Fixed Positioning: With fixed $\mathbf{q}^t$, the subproblem is to find the optimal beamformers $\mathbf{w}^t$ that maximize the sum-QoE. The required minimum SINR, $\gamma_k^{\min}$, is derived from the semantic constraint as $\gamma_k^{\min} = b_{\tau_k^t} - \frac{1}{a_{\tau_k^t}} \ln(\frac{c_{\tau_k^t}}{\varepsilon_{\mathrm{th}}} - 1)$. The beamforming problem is then to maximize the sum-QoE subject to the power constraints in (5) and the effective SINR constraints, $\gamma_k \geq \max(\gamma_{\mathrm{th}}, \gamma_k^{\min})$. Such problem is a standard SOCP and can be solved efficiently using convex optimization solvers. If the SOCP solver identifies that no feasible $\mathbf{w}^t$ exists for the current SCR profile, the continuous optimization step is declared infeasible and the corresponding QoE reward fed back to the DRL agent is set to zero, which implicitly penalizes unachievable SCR selections.

2) Positioning Optimization for Fixed Beamforming: With fixed beamformers $\mathbf{w}^t$, the objective function depends only on the UAV positions $\mathbf{q}^t$, updated via the gradient projection method (GPM). The gradient of the objective function with respect to each UAV's position, $\nabla_{\mathbf{q}_n^t} \sum_k \mathrm{QoE}_k^t$, is computed numerically. The positions are then updated along the projected gradient direction $\mathbf{q}_n^{j+1} = \mathcal{P}_{\mathcal{F}}(\mathbf{q}_n^j + \eta \nabla_{\mathbf{q}_n^j} \sum_k \mathrm{QoE}_k^t)$, where j is the AO iteration index, η is a positive step size, and $\mathcal{P}_{\mathcal{F}}(\cdot)$ is the projection operator that enforces the feasibility of the constraints in (2) and (3).

The proposed AO algorithm theoretically ensures convergence based on block coordinate descent principles. In each iteration, the beamforming subproblem is convex and solved to global optimality, while the UAV positioning subproblem employs gradient projection to obtain feasible improvements. This iterative procedure monotonically increases the sum-QoE, which is upper-bounded due to the logarithmic structure and bounded semantic similarity. Consequently, the AO process converges to a stationary point of the original problem.

Algorithm 2 Joint Positioning and Beamforming via AO

Input: SCR vector $\boldsymbol{\tau}^t$, previous positions $\mathbf{q}^{t-1}$.
Initialize: Iteration $j = 0$, initial positions $\mathbf{q}^{t,0} = \mathbf{q}^{t-1}$.
repeat
  Fix $\mathbf{q}^{t,j}$: Solve for optimal beamformers $\mathbf{w}^{t,j+1}$ via SOCP.
  Fix $\mathbf{w}^{t,j+1}$: Update positions to $\mathbf{q}^{t,j+1}$ via GPM.
  $j \leftarrow j + 1$.
until convergence or $j \geq J_{\max}$.
Output: Optimized positions $\mathbf{q}^{t*}$ and beamformers $\mathbf{w}^{t*}$.

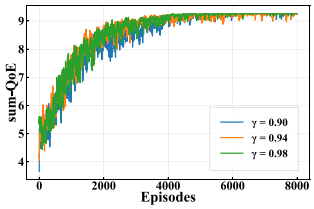


Fig. 2. Convergence of the PPO agent under different γ.

TABLE I
COMPARISON OF ALGORITHMIC COMPLEXITY AND RUNTIME

Metric	HALO	DQN-GPM	PPO-SCA
Complexity	$O((NMK)^{3.5})$	$O((NMK)^{3.5})$	$O((NMK + K^2)^{3.5})$
Training Time (s)	1552.06	1662.43	9718.75

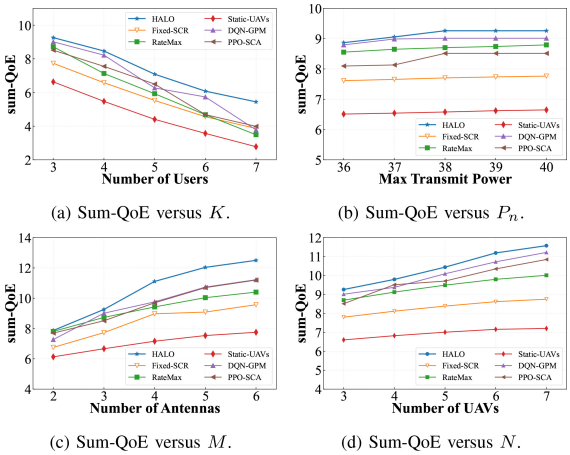
The total computational complexity per AO iteration is governed by the two subproblems. The dominant factor is the beamforming subproblem $(\mathbf{w}^t)$, which, as an SOCP, requires a complexity of $O((NMK)^{3.5})$. The UAV positioning subproblem $(\mathbf{q}^t)$ involves a lower complexity of $\mathrm{O}(N^2M^2K)$, dominated by the numerical gradient calculation.

V. Simulation Results

We consider a 1000×1000 m^2 service area with randomly distributed ground users. Default parameters involve a fleet of $N = 3$ UAVs serving $K = 3$ users, each UAV equipped with $M = 3$ antennas and maintaining a fixed altitude $H = 100$ m. Physical constraints include a minimum inter-UAV separation of $d_{\min} = 50$ m and maximum transmit power $P_n = 38$ dBm per UAV. The channel model uses a path loss exponent $\alpha = 2.8$ and a reference gain $\beta_0 = -50$ dB, with total noise power $\sigma^2 = -96$ dBm and a 20 MHz bandwidth. For the QoE model, parameters are set to $\lambda_1 = 8$ and $\lambda_2 = 1$ [11]. The PPO agent utilizes two 256-neuron FC layers, trained for 8000 episodes using the Adam optimizer, with learning rate 10^{-4}, a discount factor $\gamma = 0.98$, and clipping parameter $\epsilon = 0.2$.

To demonstrate the superiority of our proposed HALO algorithm, we compare its performance against five benchmarks: **Fixed-SCR:** This scheme eliminates the adaptive semantic compression by assigning a fixed, medium SCR. The UAV positioning and cooperative beamforming are then jointly optimized [11]. This serves to quantify the performance

Authorized licensed use limited to: Chung-ang Univ. Downloaded on October 27,2026 at 10:36:29 UTC from IEEE Xplore. Restrictions apply.



(a) Sum-QoE versus K. (b) Sum-QoE versus P_n.

(c) Sum-QoE versus M. (d) Sum-QoE versus N.

Fig. 3. Performance comparison under varying parameters.

gain attributable to the adaptive SCR selection in HALO. **Rate-Maximization (RateMax):** This is a conventional, semantic-agnostic scheme that aims to maximize the sum-rate, a common objective in traditional wireless networks [7]. This fixes the SCR to the lowest level to transmit maximum data, highlighting the crucial benefit of adopting a QoE-aware objective over a rate-centric one. **Static-UAVs:** The UAVs are placed at fixed locations. Only the SCRs and beamformers are optimized [9]. Comparing HALO to this benchmark reveals the significant performance contribution of dynamic UAV positioning optimization. **DQN-GPM:** The GPM – based AO framework for continuous variables is retained, while the PPO agent is replaced with a Deep Q-Network (DQN) for SCR selection. Such a setup validates the superiority of the PPO policy-gradient method over the value-based DQN in optimizing discrete SCR decisions. **PPO-SCA:** This variant preserves the PPO agent but replaces GPM with the Successive Convex Approximation (SCA) technique. SCA solves the joint continuous problem by iteratively linearizing its non-convex constraints, which quantifies the gains in computational efficiency and stability achieved by selecting GPM over SCA.

First, we examine the convergence of the PPO agent in our HALO algorithm. Fig. 2 plots the episodic reward (sum-QoE), which converges stably after approximately 4000 episodes for different discount factors and represents an improvement of over 130% from initial performance, confirming the stability and effectiveness of our DRL-based approach for the complex discrete resource allocation task. Crucially, the computational analysis in Table I, conducted under identical experimental settings, further confirms that the HALO framework maintains the lowest overall complexity and the fastest training runtime among all comparable iterative optimization schemes.

Fig. 3 illustrates the superiority of the HALO algorithm across varying parameters. Specifically, Fig. 3(a) shows that despite interference increasing with the number of users, HALO outperforms benchmarks by intelligently managing resources to find the best possible QoE trade-off. In Fig. 3(b), HALO remains superior under increased max transmit power because it optimizes both beamforming and semantic compression, yielding greater QoE gains than power-only strategies. Fig. 3(c) confirms that HALO most effectively translates the superior beamforming gains from larger antenna arrays into maximized QoE. Finally, Fig. 3(d) demonstrates that as the number of UAVs grows, HALO maintains its decisive lead by effectively coordinating a larger fleet to leverage enhanced spatial diversity. Unless otherwise stated, all numerical results are averaged over 500 independent realizations of random user locations and small-scale fading. Overall, the HALO framework achieves an average QoE improvement of approximately 37.5% over the strongest benchmark. The consistent and widening performance gaps across all these scenarios highlight the synergistic benefits of our holistic design.

VI. Conclusion

In this letter, we developed a multi-UAV semantic communication framework that jointly optimizes UAV positioning, cooperative beamforming, and semantic compression. The problem was formulated as maximizing QoE and solved using HALO, a hybrid algorithm combining DRL with alternating optimization. Simulations demonstrated that HALO outperforms other benchmark schemes, highlighting the advantage of holistic resource management in semantic-aware UAV networks. Future work could extend HALO to scenarios with temporally varying user locations and more realistic 6G channel models, as well as explore the impact of imperfect CSI and the integration of reconfigurable intelligent surfaces.

References

[1] S. Ma et al., "Features disentangled semantic broadcast communication networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 6, pp. 6580–6594, Jun. 2024.

[2] Y. Zhao et al., "Joint content caching, service placement, and task offloading in UAV-enabled mobile edge computing networks," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 1, pp. 51–63, Jan. 2025.

[3] K. Meng et al., "UAV-enabled integrated sensing and communication: Opportunities and challenges," *IEEE Wireless Commun.*, vol. 31, no. 2, pp. 97–104, Apr. 2024.

[4] L. Yan et al., "Resource allocation for text semantic communications," 2022, *arXiv:2201.06023*.

[5] J. Devlin et al., "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. North Am. Chapter Assoc. Comput. Linguist., Human Lang. Tech. (NAACL-HLT)*, 2019, pp. 4171–4186.

[6] X. Mu et al., "Heterogeneous semantic and bit communications: A semi-NOMA scheme," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 155–169, Jan. 2023.

[7] H. Yu et al., "Aggregate throughput Maximization for UAV-enabled relay networks with wireless power transfer: Joint trajectory and power optimization," *IEEE Trans. Veh. Technol.*, vol. 73, no. 6, pp. 8253–8265, Jun. 2024.

[8] W. Chen et al., "Adaptive hybrid beamforming for UAV mmWave communications against asymmetric jitter," *IEEE Trans. Wireless Commun.*, vol. 23, no. 8, pp. 9432–9445, Aug. 2024.

[9] X. Ji and T. Wang, "Energy minimization for fixed-wing UAV assisted full-duplex relaying with bank angle constraint," *IEEE Wireless Commun. Lett.*, vol. 12, no. 7, pp. 1199–1203, Jul. 2023.

[10] L. Li et al., "Network state sensing-assisted resource allocation for grant-free multiple access with users of heterogeneous delay tolerance," *IEEE Trans. Veh. Technol.*, vol. 74, no. 12, pp. 19366–19378, Dec. 2025.

[11] C. Zhan and R. Huang, "Energy efficient adaptive video streaming with rotary-wing UAV," *IEEE Trans. Veh. Technol.*, vol. 69, no. 7, pp. 8040–8044, Jul. 2020.

[12] W. Wu et al., "Goal-oriented semantic communications for robotic waypoint transmission: The value and age of information approach," *IEEE Trans. Wireless Commun.*, vol. 23, no. 12, pp. 18903–18915, Dec. 2024.

Authorized licensed use limited to: Chung-ang Univ. Downloaded on October 27,2026 at 10:36:29 UTC from IEEE Xplore. Restrictions apply.